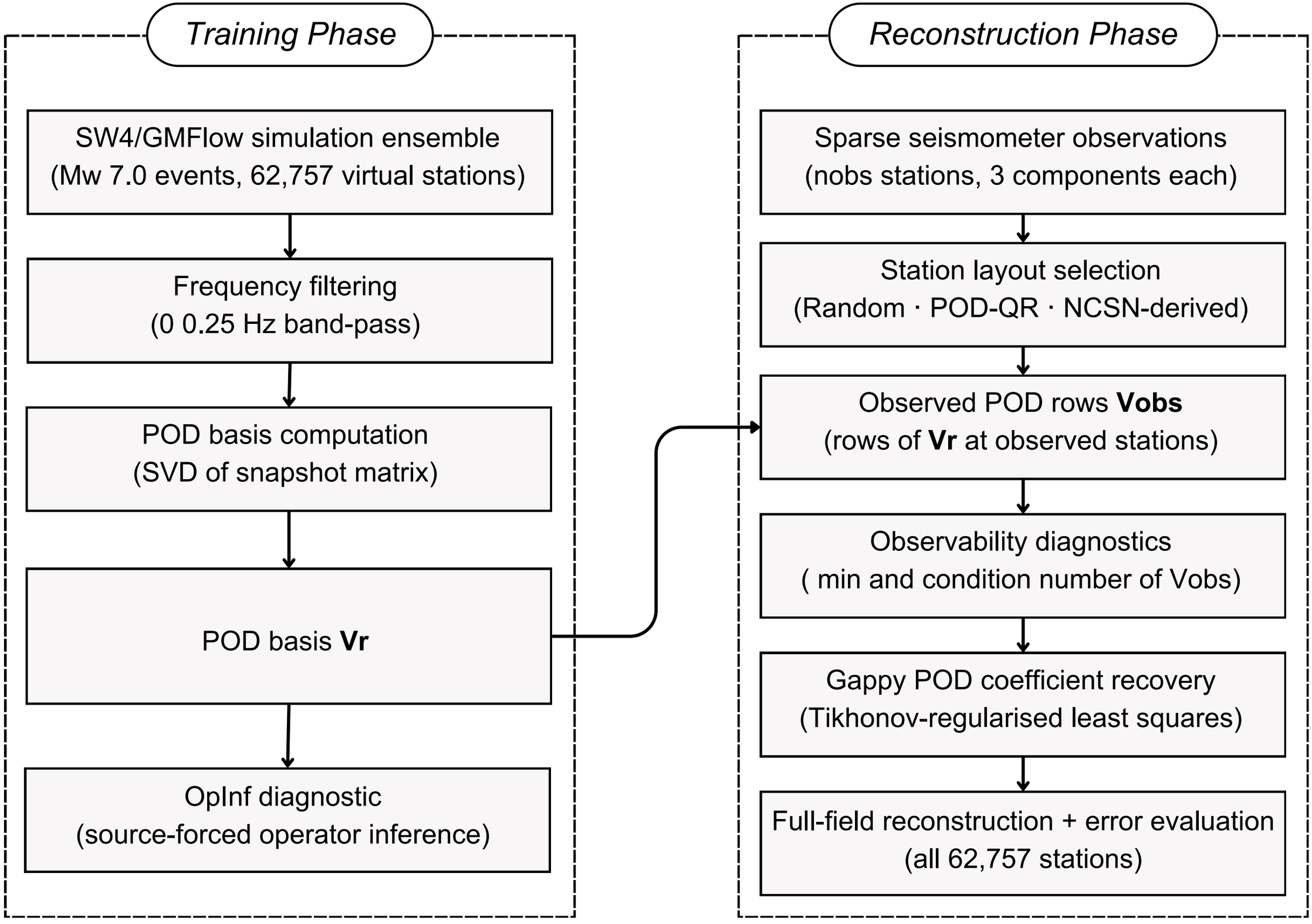

Training Phase
SW4/GMFlow simulation ensemble
(Mw 7.0 events, 62,757 virtual stations)
Frequency filtering
(0 0.25 Hz band-pass)
POD basis computation
(SVD of snapshot matrix)
POD basis Vr
OpInf diagnostic
(source-forced operator inference)
Reconstruction Phase
Sparse seismometer observations
(nobs stations, 3 components each)
Station layout selection
(Random · POD-QR · NCSN-derived)
Observed POD rows Vobs
(rows of Vr at observed stations)
Observability diagnostics
(min and condition number of Vobs)
Gappy POD coefficient recovery
(Tikhonov-regularised least squares)
Full-field reconstruction + error evaluation
(all 62,757 stations)